

Session 3

JSP Implicit Objects



Review

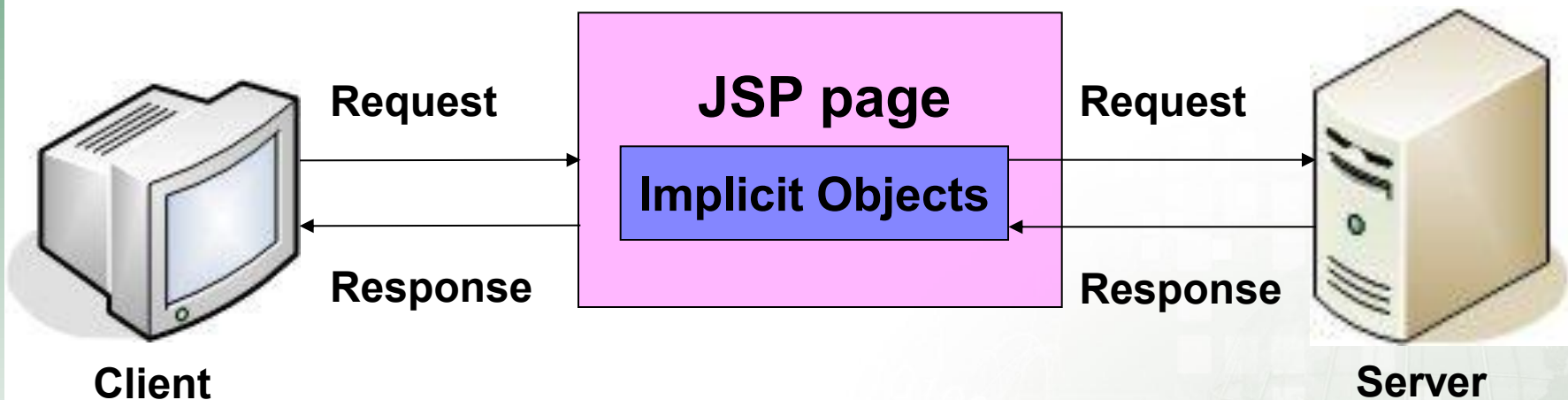
- ❑ JSP page consists of three elements: JSP expressions, JSP scriptlets and JSP declarations.
- ❑ JSP elements are enclosed within `<% =` and `%>` tags.
- ❑ JSP directives are of three types: `page`, `include`, and `taglib`.
- ❑ JSP directives are enclosed within `<%@` and `%>` tags.
- ❑ The `page` directive is used to set attributes of a JSP page.
- ❑ The `include` attribute is used to include another file in a JSP page.
- ❑ The `taglib` directive allows the user to create custom tags.

Objectives

- ❑ Explain input/output objects: `request`, `response`, `out`
- ❑ Explain control objects: `session`, `application`, `pageContext`
- ❑ Explain Servlet-Related Objects: `page`, `config`
- ❑ Explain Error Object: `exception`

What is Implicit Object?

- JSP Implicit Objects are the Java objects that the JSP Container makes available to developers in each page.

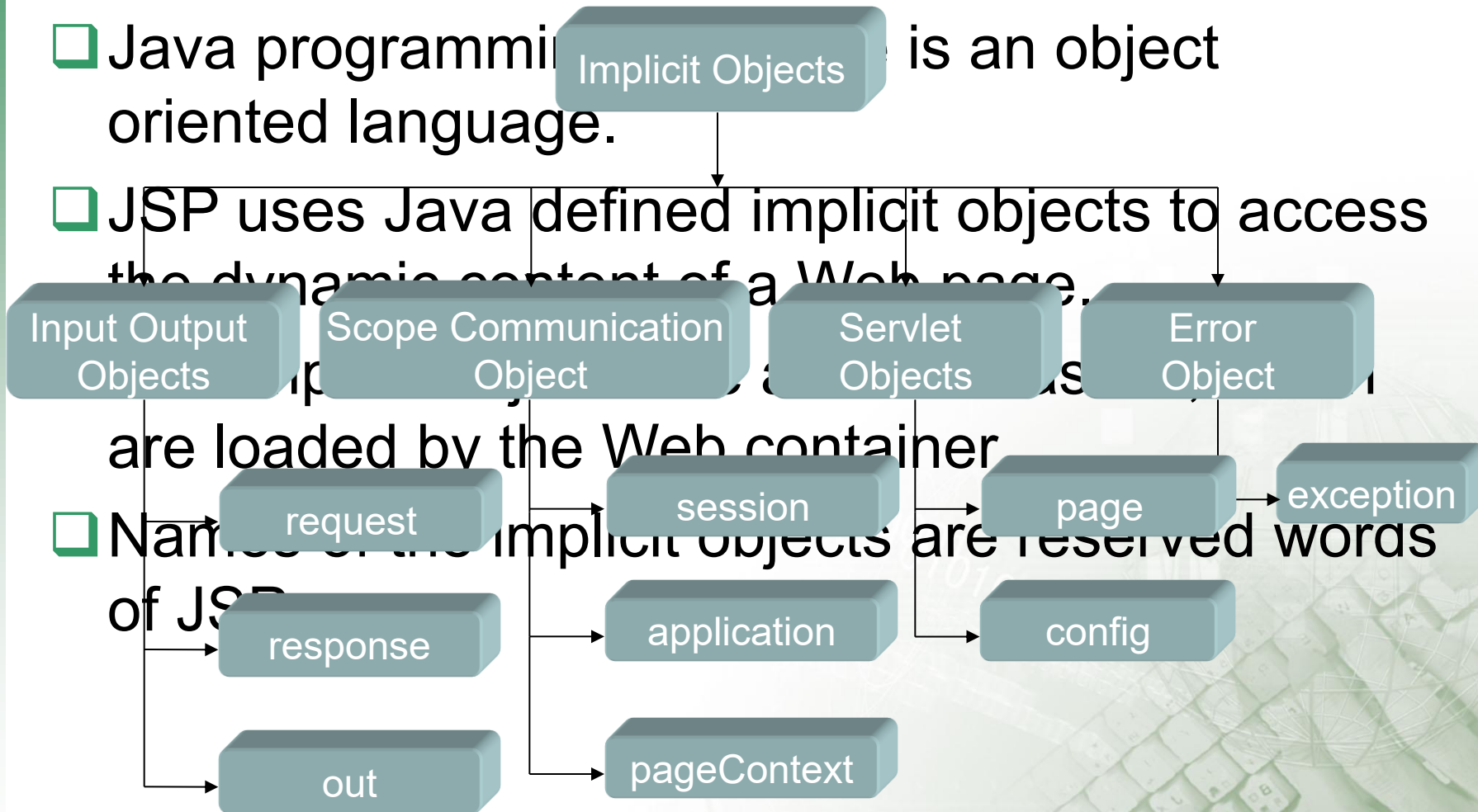


- Developers can call them directly without being explicitly declared.
- JSP Implicit Objects are also called pre-defined variables.

Implicit Objects

- Java programming is an object oriented language.

- JSP uses Java defined implicit objects to access the dynamic content of a Web page.



Input and Output Objects

- ❑ Control page input and output
- ❑ Access the data related to all the request and responses
- ❑ Input and output objects are `request`, `response`, and `out`.

request Object 2-1

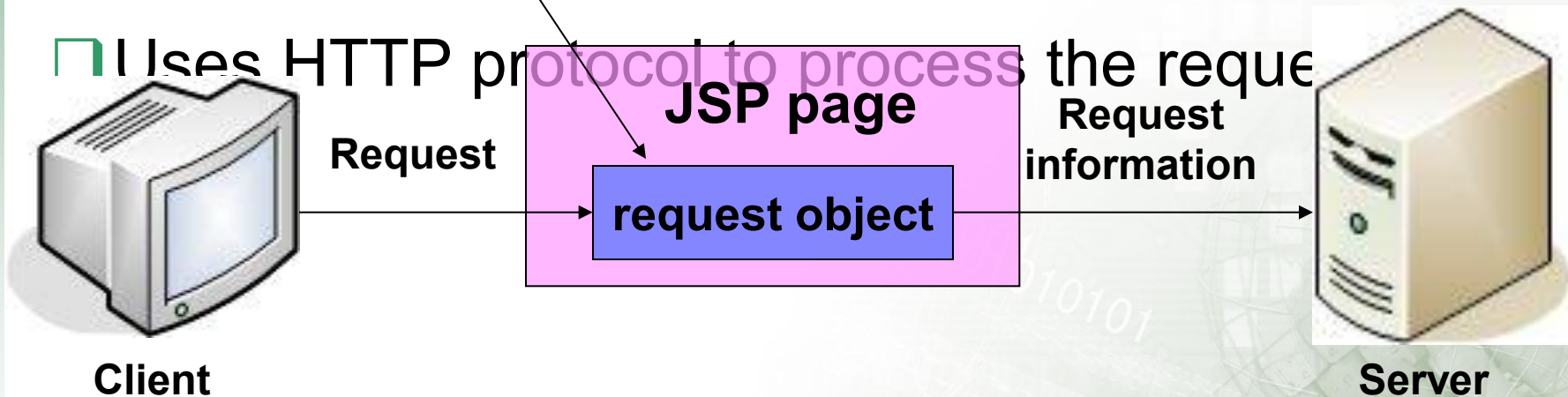
- ❑ Represents the request from the client for a Web page.

- ❑ Implem

Represents and stores the request from the client

`javax.servlet.ServletException` interface.

- ❑ Uses HTTP protocol to process the request



request Object 2-2

- ❑ Uses `getParameter()` method to access the values of parameter.

```
String xStr =  
request.getParameter("numtest");
```

try

{

long num =

Long.parseLong(xStr);

}

%>

Factorial result:

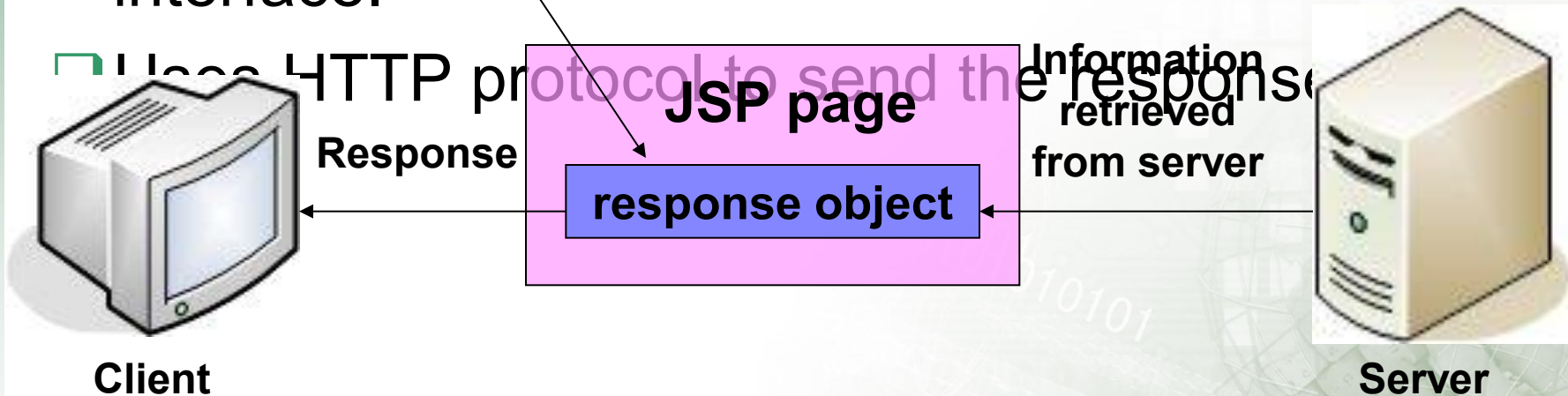
<%= num %>! = <%= fact(num) %>

request
object

response Object 3-1

- ❑ Handles the response generated by JSP.
- ❑ Sends information to the client.
- ❑ Implements the `javax.servlet.ServletResponse` interface.

Stores information to send as a response to Client request



response Object 3-2

```
<html>
  <head>
    <title>Implicit Object</title>
  </head>
  <body>
    <h1>Implicit Object</h1>
    <%
      response.setDateHeader("Expires", 0);
      response.setHeader("Pragma", "no-cache");
      if (request.getProtocol().equals("HTTP/1.1")) {
        response.setHeader("Cache-Control", "no-
cache");
        out.println("<b>Protocol used is:</b> " +
          request.getProtocol());
      }
    %>
  </body>
</html>
```

response object
sets header values

response Object 3-3

```
<%
```

```
    Cookie[] allCookies = request.getCookies();
```

```
    Cookie ourCookie = null;
```

```
<%
```

```
    } else {
```

```
%>
```

The following cookie was added earlier to your machine:

```
<br>Version: <%=ourCookie.getVersion() %>
```

```
<br>Name: <%=ourCookie.getName() %>
```

```
<br>Value: <%=ourCookie.getValue() %>
```

```
<br>MaxAge: <%=ourCookie.getMaxAge() %>
```

```
<%
```

```
}
```

```
%>
```

```
    response.addCookie(cookie);
```

```
%>
```

A cookie has been added to your machine!

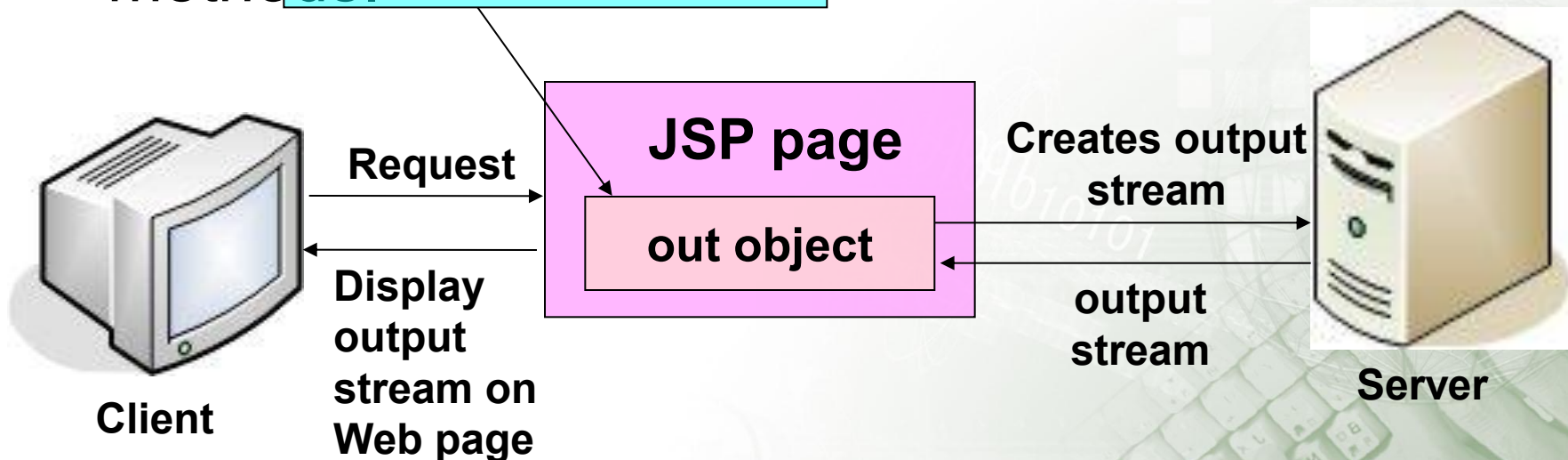
```
<br>Select refresh to see the details.
```

response object
used to add cookie

out Object 2-1

- ❑ Represents the output stream.
- ❑ Instance of `javax.servlet.jsp.JspWriter` class.
- ❑ Uses `write()`, `print()`, and `println()` methods.

Creates output stream to display information



out Object 2-2

☐ Uses `write()` method to display the output on the Web page.

Out object used to retrieve buffer size

☐ Uses `getBufferSize()` and `getRemaining()` to retrieve the buffer size.

```
<html>
<head>
<title>Buffer Size of Implicit Out</title>
</head>
<body>
<h1>Buffer Size of Implicit Object</h1>
<%
    int size = out.getBufferSize();
    int availsize = out.getRemaining();
    int usedsize = size - availsize;
%>
<b>Buffering Status:</b><br/>
<b>Total Size: </b><%=size %><br/>
<b>Available Size: </b><%=availsize %><br/>
<b>Used Size: </b><%=usedsize %><br/>
</body>
</html>
```

Control Objects

- ❑ Provide access to all objects available in the given scope.
- ❑ Objects in a JSP application are accessible according to the specified scope.
- ❑ A scope of an object is the section where that object is accessible.
- ❑ Scopes for implicit objects are: page, request, session, and application.
- ❑ Control objects are `session`, `application`, and `pageContext`.

session Object

- ❑ Web server creates a session for multiple requests sent by a single user

```
<%
    UserLogin userLoginObj1 =
        (UserLogin)session.getAttribute("loginuser");
    if (userLoginObj1.isGroupMember("admin"))
    {
        session.setMaxInactiveInterval(60*60*4);
    }
    else
    {
        session.setMaxInactiveInterval(60*5);
    }
%>
```

The diagram illustrates the session object and its usage in the provided code snippet. A purple oval labeled "session object" has arrows pointing to the `session` variable in the code. A blue oval labeled "Defined in previous code snippet" points to the `session.setMaxInactiveInterval` calls. Red dashed boxes highlight the `session` variable and the `session.setMaxInactiveInterval` calls.

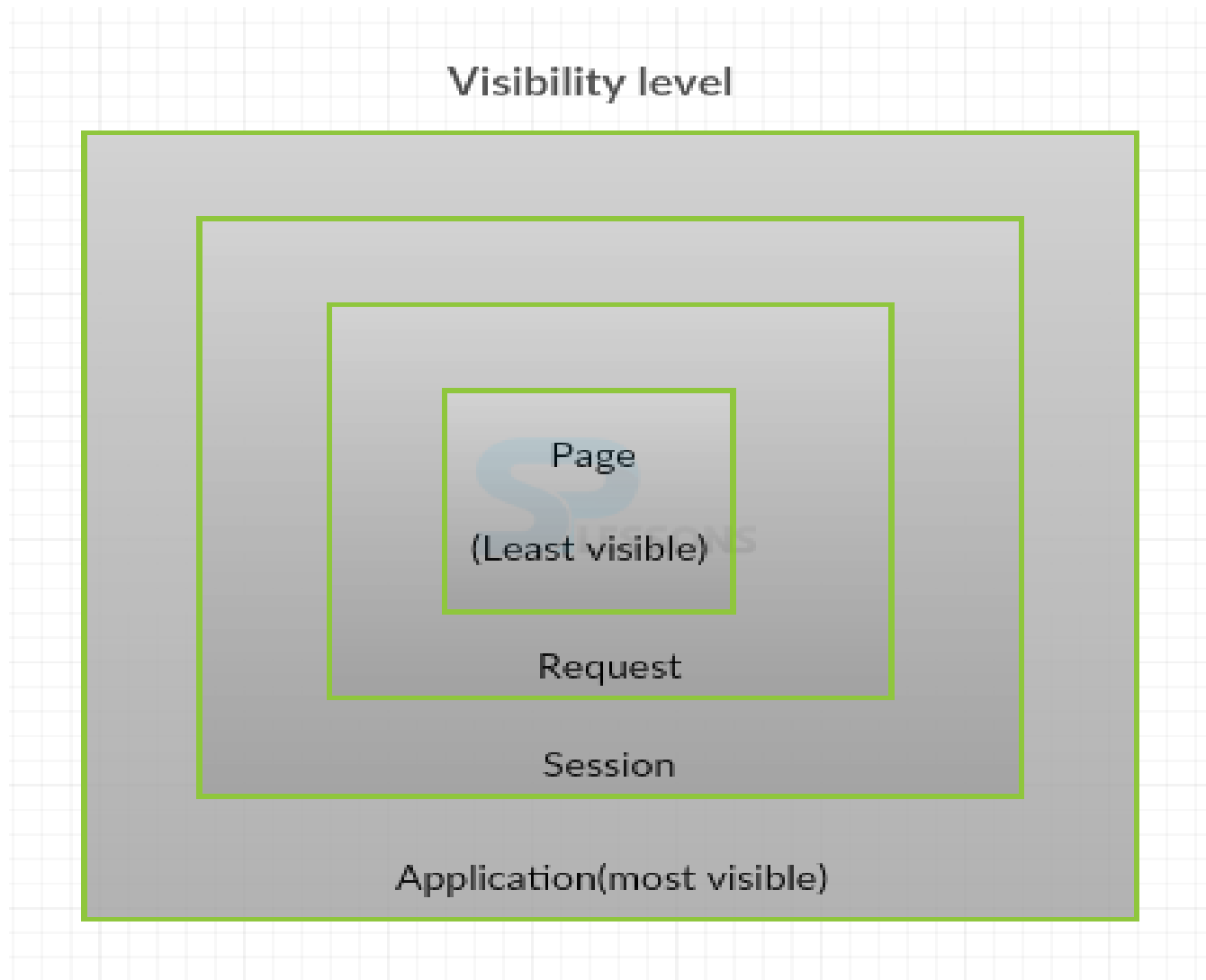
application Object

- ❑ Represents the application to which the JSP page belongs.
- ❑ JSP pages of an application are grouped together and form an application.
- ❑ Instance of `javax.servlet.ServletContext` interface.

Application Object

- ❑ The JSP Application Object is one of the implicit object type `ServletContext`
- ❑ It is created only once by the web application when the application is deployed on the web apps in Tomcat server.
- ❑ The JSP Application Object is used to get the information from the `web.xml` as `context-param` tag inside `init-param`.

Visibility Level



pageContext Object 2-1

- ❑ Allows user to access all the implicit objects defined in the page scope.
- ❑ JSP Page Context Object is used to store and retrieve the page-related information and sharing objects.
- ❑ Provides methods to access all attributes defined by implicit objects in a page.
- ❑ Also provides methods to transfer the control from one Web page to another.
- ❑ Instance of the class
`javax.servlet.jsp.PageContext.`

pageContext Object 2-2

pageContext

Object uses
getAttribute()
method

method

```
<%@ page language="java" %>
<html>
  <body>
    <h1>session.application.pageContext</h1>
    <b>Page Count = </b>
    <%=pageContext.getAttribute("pageCount") %>
    <br/><b>Session count = </b>
    <%=session.getAttribute("sessionCount") %>
    <br/><b>Application count = </b>
    <%=application.getAttribute("appCount") %>
    <br/><b>Time = </b>
    <%=newjava.sql.Time(System.currentTimeMillis()) %>
  </body>
</html>

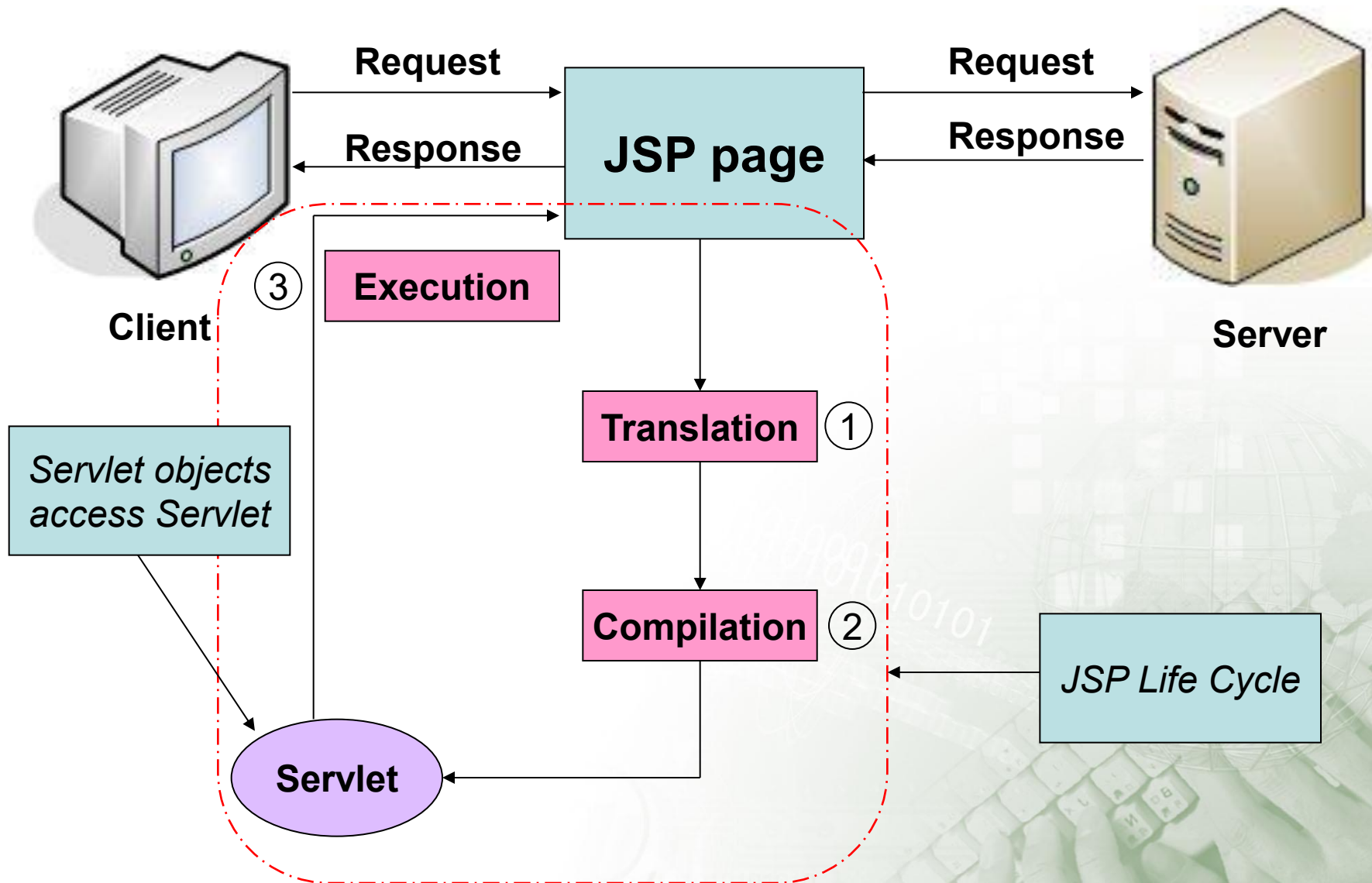
(Integer) application.getAttribute("appCount");
application.setAttribute("appCount", new
Integer(count3.intValue()+1));

%>
```

Servlet Objects 2-1

- ❑ JSP engine generates a Servlet for each JSP.
- ❑ Provide methods and variables to access information of Servlet.
- ❑ Servlet objects are:
 - ❑ page
 - ❑ config

Servlet Objects 2-2



page Object

- ❑ Provides access to all the objects which are defined in a Web page.
- ❑ Represents the page itself.
- ❑ Using `page` object, all variables and methods of Servlet class can be accessed.
- ❑ Instance of the `javax.servlet.jsp.JspPage` interface.

config Object

- ❑ Stores the information of Servlet, which is created during the compilation of a JSP page.

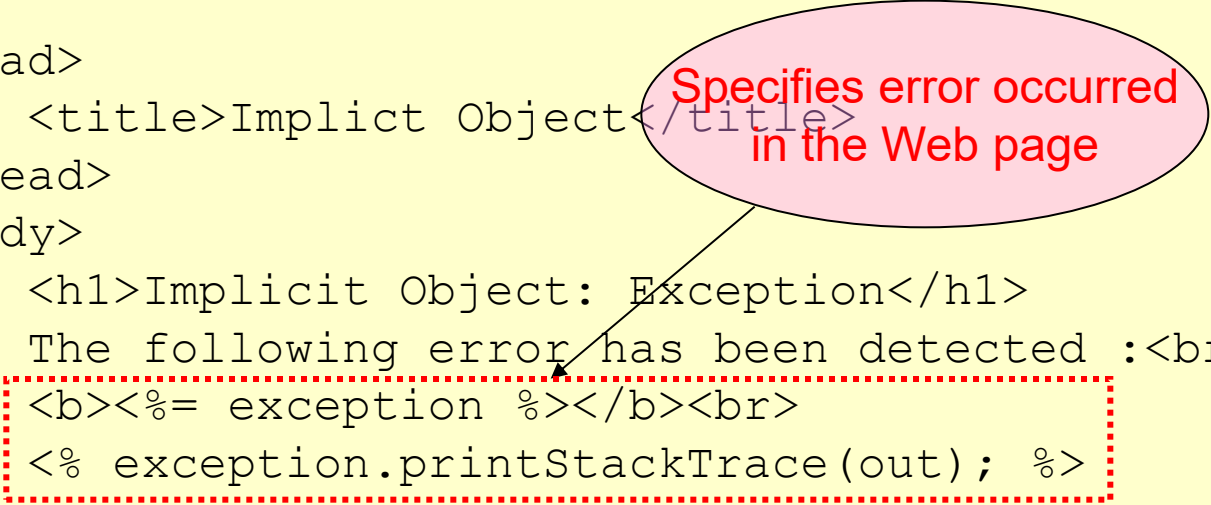
```
while( initParams.hasMoreElements() )
{
    String initParam =
    (String)initParams.nextElement();
    context.log(initParam);
    context.log(config.getInitParameter(initParam));
}
```

Config object
initializes
parameters

error Object 2-1

- ❑ Error objects are used to handle errors in a JSP page

```
<%@ page isErrorPage="true" %>
<html>
  <head>
    <title>Implicit Object</title>
  </head>
  <body>
    <h1>Implicit Object: Exception</h1>
    The following error has been detected :<br>
    <b><%= exception %></b><br>
    <% exception.printStackTrace(out); %>
  </body>
</html>
```



error Object 2-2

```
<%@ page isErrorPage="true" import="java.io.*" %>

<% if ( exception != null ) {
    out.write("\nAn error occurred\n");
} else {
    out.write("\nYou have reached this page, but no error
information is available.\n");
}
%>

<%
    out.println("<!--");
    StringWriter sw = new StringWriter();
    PrintWriter pw = new PrintWriter(sw);
    exception.printStackTrace(pw);
    out.print(sw);
    sw.close();
    pw.close();
    out.println("-->");
%>
```

Displays message if error occurred as checked exceptions are thrown

Prints the message on the JSP page

Demonstration: Example 5 and 6

Summary 2- 1

- ❑ Objects are basic building blocks of the program that consists of data and the methods to control data.
- ❑ JSP uses java programming language to access the dynamic content of a Web page with the help of java defined implicit objects. JSP implicit objects are a set of classes which are loaded by Web container.
- ❑ The JSP defined scopes are: page, request, session, and application.
- ❑ Implicit objects in JSP include: page, config, request, response, out, session, application, pageContext, and exception.

Summary 2-2

- ❑ The HTTP protocol is used to process the request from the client. The request object also implements the `javax.servlet.ServletRequest` interface.
- ❑ The response object implements the `javax.servlet.ServletResponse` interface.
- ❑ The out object is the output stream for the Web page. The out object is an instance of the `javax.servlet.jsp.JspWriter` class.